

Problem set 9

Page Rank Algorithm

Exercise 1

Consider the following dataset:

$$A = (0, 2), \quad B = (0, 1), \quad C = (1, 0).$$

The k -means algorithm is initialized with centers at A and B . Upon convergence, what will be the two centers of the clusters?

Exercise 2

Consider the problem of binary classification using the Naive Bayes classifier. You are given two dimensional features (X_1, X_2) and the categorical class conditional distributions in the tables below. The entries in the tables correspond to $P(X_1 = x_1 \mid C_i)$ and $P(X_2 = x_2 \mid C_i)$ respectively. The two classes are *equally likely*. Given a data point $(1, -1)$, calculate $P(C_1 \mid X_1 = -1, X_2 = 1)$.

$X_1 =$ \ Class	C_1	C_2
-1	0.2	0.3
0	0.4	0.6
1	0.4	0.1

$X_2 =$ \ Class	C_1	C_2
-1	0.4	0.1
0	0.5	0.3
1	0.1	0.6

Figure 1:

Exercise 3

The table below is a distance matrix for 6 objects.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
<i>A</i>	0					
<i>B</i>	0.12	0				
<i>C</i>	0.51	0.25	0			
<i>D</i>	0.84	0.16	0.14	0		
<i>E</i>	0.28	0.77	0.70	0.45	0	
<i>F</i>	0.34	0.61	0.93	0.20	0.67	0

Table 1: Table 1

Show the final result of hierarchical clustering with single link by drawing a dendrogram.

Exercise 4

Show the final result of hierarchical clustering with complete link by drawing a dendrogram.

Exercise 5

Change two values from the Table 1 so that your answer to the last two questions would be same.

Exercise 6

Which clustering method(s) is most likely to produce the following results at $k = 2$? Choose the most likely method(s) and briefly explain why it/they will work better where others will not in **at most 3 sentences**. Here are the five clustering methods you can choose from:

- Hierarchical clustering with single link
- Hierarchical clustering with complete link
- Hierarchical clustering with average link
- k -means

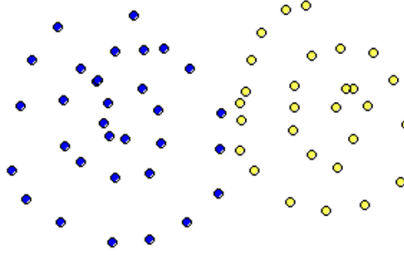


Figure 2: Data Distribution Within Clusters.

Exercise 7

Suppose that an IR system contains only 1000 documents numbered d_1 to d_{1000} . A query is known to generate 27 relevant documents as listed below:

$$\{d_1, d_5, d_7, d_{10}, d_{88}, d_{151}, d_{200}, d_{211}, d_{250}, d_{300}, d_{399}, d_{401}, d_{405}, \\ d_{450}, d_{473}, d_{500}, d_{501}, d_{530}, d_{545}, d_{590}, d_{600}, d_{735}, d_{700}, d_{720}, d_{800}, d_{888}, d_{900}\}.$$

Two very different IR systems are used to retrieve ranked documents for this query. Each system only returns the top 10 ranked documents in the following order of ranking. Systems 1 and 2 each retrieve documents one at a time in the following order with all 10 documents eventually returned:

- **System 1:** $d_{122}, d_{211}, d_{150}, d_{88}, d_{37}, d_1, d_{501}, d_{800}, d_{201}, d_5$.
- **System 2:** $d_{10}, d_6, d_{700}, d_{250}, d_{88}, d_{600}, d_{59}, d_{422}, d_{500}, d_{333}$.

Answer the following and show your work:

- Calculate the F1 score for each.
- Which IR system is better? Justify your answer.

Exercise 8

An IR system returns 8 relevant documents, and 10 nonrelevant documents. There are a total of 20 relevant documents in the collection. What is the precision of the system on this search, and what is its recall?

Exercise 9

Consider an information need for which there are 4 relevant documents in the collection. Contrast two systems run on this collection. Their top 10 results are judged for relevance as follows (the leftmost item is the top ranked search result):

• **System 1:** R N R N N N N N R R

• **System 2:** N R N N R R R N N N

1. What is the AP of each system? Which has a higher AP?
2. What is the R-precision of each system? (Does it rank the systems the same as AP?)

Exercise 10

Which of the following expressions is equivalent to $p(A \mid B, C, D)$?

(a)

$$\frac{p(A, B, C, D)}{p(C \mid B, D) p(B \mid D) p(D)}$$

(b)

$$\frac{p(A, B, C, D)}{p(B, C) p(D)}$$

(c)

$$\frac{p(A, B, C, D)}{p(B, C \mid D) p(B) p(C)}$$